

Supporting Information for “Closed-loop, event-driven machine learning for autonomous triage of single-photon emitters”

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S1 Simulator details and validation

S1.1 Exact reference

The three-level rate matrix $M(k_{exc}, k_r, k_{es}, k_{se})$ has one zero eigenvalue (the steady state) and two negative eigenvalues; the conditional excited-state population $p_e(\tau|g)$ is computed by eigen-decomposition, so the reference $g^{(2)}(\tau) = p_e(\tau|g)/p_e^{ss}$ is numerically exact. The simulator maps (τ_1, τ_2, a) to rates as $k_{exc} = 0.4/\tau_1$, $k_r = 0.6/\tau_1$, $k_{se} = 1/\tau_2$, and $k_{es} = a k_{se}(k_{exc} + k_r)/k_{exc}$; the validation always compares the photon-stream simulation with the exact eigen-solution for the *same* rates, so the comparison does not depend on the accuracy of this mapping. Table S1 gives the statistics for the three parameter sets of main-text Figure 2(a)–(c). The photon-stream simulator samples every excitation and emission cycle (block-vectorized continuous-time Monte Carlo), thins the emissions to the target detected rate, applies telegraph blinking gates, adds uniform Poisson background, splits the photons 50:50, and passes them through a detector model with Gaussian timing jitter ($\sigma_{IRF} = 0.35$ ns) and, when enabled, 45-ns dead time and 2% afterpulsing with an 80-ns exponential delay.

S1.2 Histogram-simulator dispersion

Under the compound condition of main-text Figure 2(d) ($N=2$, $\rho=0.8$, IRF jitter, 500 kcps, $T=1$ s, 80 repetitions) the per-bin means of the histogram simulator agree with those of the photon-stream simulator to the percentage quoted in the main text, and the empirical Fano factor is consistent with unity.

Table S1: Photon-stream simulator compared with the numerically exact master-equation $g^{(2)}(\tau)$ (no free parameters). Effective (τ_1, τ_2, a) from the eigen-decomposition of the simulated rates; χ^2_V over the 121 delay bins with Poisson errors.

Regime	T (s)	Coinc.	τ_1 (ns)	τ_2 (ns)	a	χ^2_V	NRMSE	MAD
NV-like	3.0	119,085	13.11	140	0.87	0.856	3.9%	3.1%
hBN-like	1.0	139,673	2.83	47	1.64	1.032	5.3%	4.0%
GaN-like	0.4	135,503	1.09	40	0.52	1.211	4.1%	3.2%

S1.3 Validation over sampled and boundary parameter sets

The photon-stream versus exact comparison of the main text was repeated for 26 photophysical parameter sets (τ_1, τ_2, a) : five drawn at random from the prior of each platform (NV, hBN, GaN, SiV) and six boundary cases that combine the

extremes of the prior ranges. As in the main-text protocol, each set was simulated for a single emitter at a detected rate of 1500 kcps with $\rho = 1$ and ideal detectors, so this sweep tests the photophysical parameters only; background, multiplicity, count rate, detector effects, and blinking are tested in Table S3, Table S4, and Figure S1. The acquisition time was 1.2 s, shortened in proportion to τ_1 for the fastest emitters to keep the simulation within memory, which gave 12,251 to 250,207 coincidences per set (Table S2). The reduced χ^2 ranged from 0.78 to 1.41 (mean 1.00). The largest value, 1.41, for the shortest τ_1 combined with the longest τ_2 and the largest a , was repeated with four new noise seeds and gave $\chi^2_v = 0.83, 0.97, 0.80, 0.94$, consistent with a statistical fluctuation. We found no systematic deviation between the photon-stream simulator and the exact solution for these parameter sets; the sweep does not sample the complete joint space of all simulator parameters.

Table S3 compares the histogram simulator with the photon-stream simulator under six representative compound operating conditions that vary multiplicity, background fraction, count rate, and blinking, together with the mean absolute deviation expected from counting statistics alone for 60 repetitions of a 1-s acquisition. For the five stationary conditions the measured deviation is within a factor of 2.1 of this statistical expectation (the largest ratio occurs at the highest count rate, where the expectation is smallest), which limits any systematic difference to about 1% of the mean counts, and the Fano factors lie between 0.97 and 1.01. The blinking condition deviates by 14.4% against a statistical expectation of 1.9%, with a Fano factor of 1.15. The histogram simulator represents blinking only through the duty cycle of the telegraph process, which reproduces the mean singles rate but neither the rate fluctuations between acquisitions nor, exactly, the resulting bunching pedestal. Statements that the histogram simulator reproduces the coincidence statistics therefore apply to stationary, non-blinking sources; for blinking sources it is an approximation, and the photon-stream simulator, which models blinking event by event, was used for all validation.

Table S2: Photon-stream simulator compared with the numerically exact master-equation $g^{(2)}(\tau)$ for random and boundary photophysical parameter sets (single emitter, detected rate 1500 kcps, $\rho = 1$, ideal detectors; 121 delay bins, Poisson errors; NRMSE: normalized root-mean-square error).

Parameter set	τ_1 (ns)	τ_2 (ns)	a	Coinc.	χ^2_v	NRMSE
random-NV-0	9.81	280	0.75	112,814	0.93	0.044
random-NV-1	21.38	481	0.93	101,584	1.09	0.045
random-NV-2	22.72	336	0.73	90,154	0.85	0.038
random-NV-3	10.34	255	0.24	83,063	1.00	0.037
random-NV-4	9.04	121	0.96	115,063	1.13	0.049
random-hBN-0	4.29	271	0.87	131,134	1.03	0.050
random-hBN-1	2.02	21	0.26	85,561	0.85	0.036
random-hBN-2	4.63	21	1.94	93,878	1.02	0.041
random-hBN-3	2.54	279	2.84	228,451	1.09	0.067
random-hBN-4	1.80	797	2.45	250,207	1.11	0.072
random-GaN-0	0.61	37	0.19	13,384	1.01	0.103
random-GaN-1	1.82	67	0.65	48,631	1.04	0.066
random-GaN-2	1.62	117	0.50	43,647	1.03	0.071
random-GaN-3	1.29	95	1.04	40,759	0.92	0.082
random-GaN-4	0.64	33	0.15	13,984	0.92	0.097
random-SiV-0	1.09	180	0.74	34,177	1.03	0.091
random-SiV-1	1.84	176	0.67	55,152	0.92	0.065
random-SiV-2	1.75	77	0.77	49,424	0.83	0.063
random-SiV-3	1.82	29	0.31	40,625	0.78	0.053
random-SiV-4	1.18	154	0.77	36,613	1.03	0.089
edge-shortest-tau1-tau2	0.60	10	0.10	12,251	1.08	0.101
edge-shortest-tau1-longest-tau2-max-a	0.60	2500	3.00	47,149	1.41	0.230
edge-longest-tau1-NV-max-a	25.00	100	1.50	102,927	1.13	0.047
edge-longest-tau1-tau2-min-a	25.00	500	0.10	56,278	1.02	0.034
edge-hBN-extreme-bunching	1.50	2500	3.00	115,354	0.93	0.117
edge-GaN-fast-shelving	2.00	10	1.20	43,289	0.91	0.054

Table S3: Histogram simulator compared with the photon-stream simulator under compound operating conditions ($\tau_1 = 14$ ns, $\tau_2 = 250$ ns, $a = 0.8$; 60 repeated 1-s acquisitions each; IRF jitter; blinking condition: $t_{\text{on}} = 20$ ms, $t_{\text{off}} = 8$ ms). Measured: mean absolute relative deviation of the per-bin mean counts. Expected: the same quantity predicted by Poisson counting statistics alone for 60 repetitions.

Condition	N	ρ	Rate (kcps)	Blinking	Measured (%)	Expected (%)	Fano
N2-rho0.8-IRF	2	0.80	500	no	1.59	1.25	0.970
N1-rho0.95-IRF	1	0.95	200	no	3.64	3.10	0.981
N3-rho0.6-IRF	3	0.60	800	no	0.75	0.80	0.979
N1-rho0.7-lowrate	1	0.70	80	no	8.41	7.69	1.014
N2-rho0.9-highrate	2	0.90	1000	no	1.31	0.62	0.999
N1-rho0.85-blinking	1	0.85	400	yes	14.42	1.90	1.154

S1.4 Bin-to-bin correlations of the photon-stream simulator

The histogram simulator treats every delay bin as an independent Poisson variable. A unit Fano factor does not establish that the bins are independent, so we measured their correlations directly. For each condition, 1000 acquisitions of 0.5 s were simulated with the photon-stream simulator ($\tau_1 = 14$ ns, $\tau_2 = 250$ ns, $a = 0.8$, IRF jitter $\sigma_{\text{IRF}} = 0.35$ ns), and the 121×121 correlation matrix of the per-bin counts was computed. Condition A has two emitters, $\rho = 0.8$, and 500 kcps; condition B has one emitter, $\rho = 0.95$, and 200 kcps. Both were simulated with ideal detectors and with 45-ns dead time and 2% afterpulsing (80-ns exponential delay). Condition C is a blinking single emitter ($\rho = 0.85$, 400 kcps, $t_{\text{on}} = 20$ ms, $t_{\text{off}} = 8$ ms) with ideal detectors. Figure S1 shows the correlation matrices and Table S4 their statistics. For 1000 uncorrelated samples the expected mean absolute correlation coefficient is $\sqrt{2/\pi}/\sqrt{n_{\text{rep}}} = 0.025$, and a fraction of 4.6% of the coefficients is expected to exceed $2/\sqrt{n_{\text{rep}}}$ by chance. In the four stationary cases the mean absolute coefficient (0.025–0.026) and the fraction beyond $2/\sqrt{n_{\text{rep}}}$ (4.6–5.1%) agree with these expectations, the mean coefficient and the nearest-neighbor coefficients are close to zero, and the maps show no structure. No statistically significant bin-to-bin correlations were therefore detected under the tested stationary conditions, with or without detector dead time and afterpulsing; this does not exclude correlations below the sampling precision ($1/\sqrt{n_{\text{rep}}} = 0.032$) or under conditions that were not tested. In the blinking case, by contrast, all bins are positively correlated (mean coefficient +0.128, Fano factor 1.14) [Figure S1(e)]. The fraction of the 0.5-s acquisition that the emitter spends in the on state varies from one acquisition to the next, so the total number of coincidences fluctuates by 9.5% (relative standard deviation) instead of the 2.2% expected from Poisson statistics, and this common factor multiplies every bin. When each histogram is rescaled to the mean total count, the correlations disappear [Figure S1(f)]: the mean coefficient becomes -0.0082, equal to the value $-1/(n_{\text{bins}} - 1) = -0.0083$ imposed by the fixed total alone, and 5.4% of the coefficients exceed $2/\sqrt{n_{\text{rep}}}$. The blinking-induced correlation is therefore a fluctuation of the overall scale of the histogram rather than a coupling between particular delays. The histogram simulator, which assigns blinking emitters a fixed duty cycle, does not reproduce this extra variance of the overall scale; this is the same effect that limits it for blinking sources (Table S3).

Table S4: Statistics of the off-diagonal elements of the bin-to-bin correlation matrices of Figure S1 (1000 repetitions each; expected for uncorrelated data: mean $r = 0$, mean $|r| = 0.025$, 4.6% of coefficients with $|r| > 2/\sqrt{n_{\text{rep}}}$). Adj.: mean coefficient of adjacent bins. Last row: condition C after each histogram is rescaled to the mean total count, for which the fixed total alone gives a mean r of -0.0083.

Condition	Counts/bin	Mean r	Mean $ r $	p95 $ r $	Max $ r $	Adj.	$> 2\sigma$ (%)	Fano
A, ideal	34.3	-0.0002	0.0258	0.064	0.133	-0.0055	5.1	1.001
A, dead time + afterp.	34.9	-0.0003	0.0250	0.063	0.144	-0.0015	4.9	0.992
B, ideal	6.4	+0.0006	0.0252	0.063	0.120	+0.0052	5.0	1.002
B, dead time + afterp.	6.6	+0.0002	0.0251	0.062	0.124	+0.0049	4.6	0.999
C (blinking), ideal	17.6	+0.1282	0.1283	0.188	0.258	+0.1276	94.3	1.140
C, rescaled to mean total	17.6	-0.0082	0.0260	0.064	0.143	-0.0118	5.4	n/a

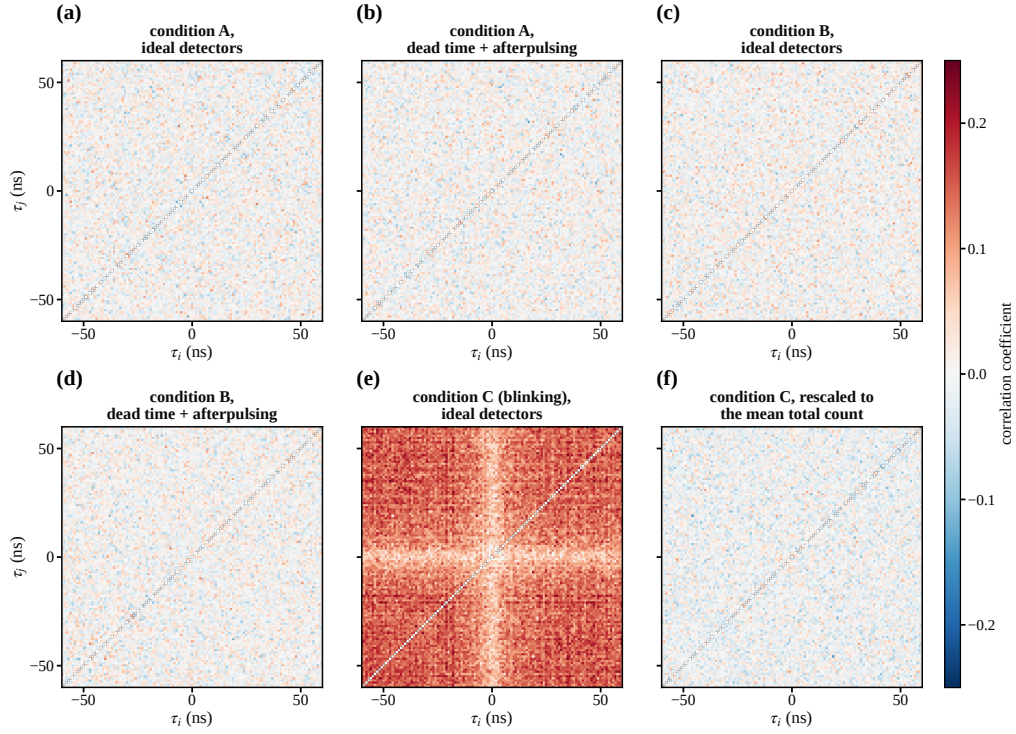


Figure S1: Correlation matrices of the per-bin coincidence counts of 1000 repeated 0.5-s photon-stream acquisitions (diagonal omitted; color scale ± 0.25). (a),(b) Condition A (two emitters, $\rho = 0.8$, 500 kcps) with ideal detectors and with dead time and afterpulsing. (c),(d) Condition B (one emitter, $\rho = 0.95$, 200 kcps), same detector settings. (e) Condition C, a blinking single emitter with ideal detectors. (f) Condition C after each histogram is rescaled to the mean total coincidence count.

S2 Platform priors

Table S5 lists the sampled parameter ranges per platform. Within a platform the antibunching time, bunching time, and count rate are log-uniform; the bunching amplitude is uniform; the signal fraction follows the 70/30 mixture described in the caption; and the blinking on and off times are log-uniform in the quoted ranges.

Table S5: Platform parameter priors (ranges anchored to the published photophysics cited in the main text). ρ is sampled from a 70/30 bimodal mixture over its range (localized-emitter vs. high-background spots); multiplicity $N \in \{1, 2, 3, 4\}$ with probabilities (0.42, 0.30, 0.18, 0.10).

Platform	τ_1 (ns)	τ_2 (ns)	a	rate (kcps)	ρ	$P(\text{blink})$
NV	8–25	100–500	0.1–1.5	30–350	0.55–0.98	0.15
hBN	1.5–5	20–2500	0.2–3.0	60–900	0.6–0.99	0.35
GaN	0.6–2.0	10–300	0.1–1.2	100–1000	0.6–0.98	0.10
SiV	0.8–2.0	15–250	0.1–0.8	80–800	0.65–0.99	0.08

S3 Monte-Carlo Bayes reference and its dependence on the reference-sample size

The Monte-Carlo Bayes reference is the decision obtained from the posterior under the same prior that the estimators are trained on: for each evaluation acquisition, exact Poisson likelihoods of the observed histogram and singles count are evaluated for M sites drawn from the prior, and the site is classified by posterior mass. It is the Bayes-optimal decision under the assumed simulation prior, not a universal physical limit. Because the posterior is computed from a finite sample of the prior, the reference accuracy is expected to increase with M , and more so at long acquisition times, where the likelihood is concentrated on a smaller fraction of the sample. Table S6 lists the reference accuracy for $M = 1500$ to 48000, averaged over 5 independent reference samples, for the fixed 1200-site NV evaluation population and noise seed of the main text. Between $M = 6000$ and 48000 the mean changes by at most 1.2 points at $T \leq 0.3$ s (at most 0.3 points between $M = 24000$ and 48000); at $T = 1$ s it increases by 2.3 points, of which 0.4 between $M = 24000$ and 48000; and at $T = 30$ s it increases by 4.5 points, of which 0.9 in the last doubling. At $T = 30$ s the reference is therefore still increasing with M over the tested range, which is why trained estimators can exceed the finite- M value at long acquisition times; we use the reference only for $T \leq 1$ s. The main-text Figure 3(a) shows the $M = 48000$ mean. The single-sample $M = 6000$ values listed in Table S9 (last row), which were plotted in the previous versions of Figure 3(a), are lower by up to 2.3 points at $T \leq 1$ s.

Table S6: Monte-Carlo Bayes reference (balanced accuracy, %) versus reference-sample size M : mean $\pm$ standard deviation over 5 independent reference samples; fixed 1200-site NV evaluation population and noise seed.

T	$M=1500$	$M=3000$	$M=6000$	$M=12000$	$M=24000$	$M=48000$
0.03 s	61.8 ± 1.2	63.1 ± 0.3	62.8 ± 0.3	63.1 ± 0.4	63.1 ± 0.4	62.8 ± 0.6
0.1 s	68.8 ± 0.6	70.1 ± 0.5	70.9 ± 0.7	71.4 ± 0.7	71.5 ± 0.5	71.5 ± 0.3
0.3 s	75.2 ± 0.5	77.5 ± 0.6	78.6 ± 0.5	79.3 ± 0.5	79.6 ± 0.5	79.8 ± 0.1
1 s	81.5 ± 0.6	84.7 ± 1.4	86.4 ± 0.9	87.7 ± 0.8	88.3 ± 0.4	88.7 ± 0.4
30 s	85.6 ± 0.9	89.8 ± 0.7	92.9 ± 0.2	95.2 ± 0.6	96.5 ± 0.7	97.4 ± 0.3

S4 Robustness to prior misspecification

The estimators are trained on the stated NV prior, which a real sample will not match exactly. To test the sensitivity, the trained SNN and CNN were evaluated without retraining on populations of 1200 sites drawn from deliberately shifted priors: the antibunching-time range scaled by $\pm 30\%$, the count-rate range scaled by 0.6, the bunching-time range scaled by 1.5, and the blinking probability doubled. For each prior, balanced accuracy was computed at all seven acquisition times of the main text with five noise seeds for the networks and two noise seeds (400 sites) for the LM fit, as in the main-text protocol, and the time to reach the 90% target was interpolated in $\log T$. Table S7 gives the balanced accuracy at 0.1 and 1 s, and Table S8 the times to target and the resulting acquisition-time ratios relative to the LM fit. Because each shifted prior also generates a different population of sites, the changes combine prior misspecification with a change in the intrinsic difficulty of the population. Lengthening the antibunching-time or bunching-time range changed the network accuracy at 0.1 and 1 s by at most 1.2 points; shortening the antibunching-time range or doubling the blinking probability reduced it by up to 3.2 points; and the reduced count-rate range reduced it by up to 7.7 points. The LM fit, which does not use the prior, also lost accuracy at the reduced count rate (74.1% versus 79.5% at 1 s), so part of this decrease reflects the smaller number of coincidences per acquisition rather than misspecification alone. Under every shifted prior the SNN reached the target 2.3 to 8.7 times faster than the LM fit (in-prior control 5.9; smallest ratio for the τ_1 range shortened by 30%). The acquisition-time reduction is therefore not specific to the training prior within the range of shifts tested, although its size depends on the prior. These values are based on one evaluation population per prior and five noise seeds for the networks (two for the LM fit). These results cover the listed shifts only. The experimental quantum-dot data of the main text differ from the CW training priors in kind (pulsed excitation and a different emitter class) and were analyzed with a separately trained network; we do not quantify the size of that mismatch.

Table S7: Balanced accuracy (% , mean $\pm$ s.d. over five noise seeds) of the trained estimators on deliberately shifted evaluation priors (no retraining; 1200 sites per prior).

Evaluation prior	SNN		CNN	
	$T=0.1$ s	$T=1$ s	$T=0.1$ s	$T=1$ s
in-prior (control)	71.1 ± 1.2	88.0 ± 1.0	71.0 ± 1.0	88.8 ± 0.9
τ_1 range +30%	71.6 ± 1.1	89.1 ± 0.7	72.2 ± 0.9	90.0 ± 0.7
τ_1 range -30%	69.1 ± 0.9	84.8 ± 1.1	68.9 ± 1.5	85.7 ± 1.2
count-rate range $\times 0.6$	63.4 ± 0.9	81.0 ± 1.1	63.7 ± 1.1	81.9 ± 1.4
τ_2 range +50%	70.9 ± 1.1	88.7 ± 1.4	71.0 ± 0.6	89.0 ± 1.3
blinking probability $\times 2$	70.0 ± 1.1	86.2 ± 0.5	70.6 ± 0.9	86.9 ± 1.2

Table S8: Acquisition time (s) to reach 90% balanced accuracy on each evaluation prior, and the ratio of the LM-fit time to the network time (n.r.: not reached within 30 s; n/a: not defined).

Evaluation prior	LM fit	CNN	SNN	Ratio (CNN)	Ratio (SNN)
in-prior (control)	8.47	1.27	1.44	6.7	5.9
τ_1 range +30%	10.45	1.01	1.21	10.4	8.7
τ_1 range -30%	6.57	2.35	2.81	2.8	2.3
count-rate range $\times 0.6$	16.60	3.44	3.87	4.8	4.3
τ_2 range +50%	11.36	1.28	1.37	8.9	8.3
blinking probability $\times 2$	9.47	1.94	2.28	4.9	4.1

S5 Detailed methods

S5.1 Estimators and training

Three estimators receive one acquisition. The *LM fit* divides the histogram by the uncorrelated coincidence level computed from the measured singles rate $\hat{r}$, $(\hat{r}/2)^2 \Delta\tau T$ per bin, and fits the two-exponential model of main-text Eq. (1) with a free overall scale factor (initialized at 1) using the Levenberg–Marquardt algorithm from four starting points; $g^{(2)}(0)$ is read relative to the fitted large-delay limit of the model. The *CNN* is a one-dimensional convolutional network on the log-scaled histogram with auxiliary scalar inputs and outputs for classification and $g^{(2)}(0)$ regression, similar to the network of Kudyshev *et al.* The *SNN* is a two-hidden-layer network of leaky integrate-and-fire neurons trained with surrogate gradients; it receives the acquisition as $S=32$ time slices of coincidence counts and returns a probability after every slice, so the time at which a decision is taken can be chosen after training.

For variable-exposure training, every batch draws fresh emitters, an acquisition time $T \sim \log \mathcal{U}[0.03, 30]$ s, and Poisson noise, so the networks see the full range from a few coincidences to well-resolved histograms; the long-exposure comparison trains the same CNN on $T = 30$ s only. Evaluation uses a fixed 1200-site population with five noise seeds (two seeds and 400 sites for the LM fit, which is the slowest estimator); $\pm$ values are standard deviations over seeds; regression quality is the mean absolute error (MAE) of the $g^{(2)}(0)$ estimate; and classification metrics exclude the boundary band $0.4 < g^{(2)}(0) < 0.6$ ($\sim 19\%$ of sites), which is covered by the regression output.

S5.2 Normalization

No estimator uses the edge of the correlation window to normalize the histogram. The uncorrelated level used by the LM fit and supplied to the networks is computed from the singles rate; the networks receive the raw counts together with five auxiliary inputs ($\log T$, $\log \hat{r}$, $\log$ total counts, $\log$ of the singles-derived uncorrelated counts in the window, and $\log$ counts within 12 ns of zero delay), and they are trained against the physical label $1 - \rho^2/N$. The same construction is used for the optimized correlation window, for which the uncorrelated level is recomputed with the corresponding bin width. On the NV prior the bunching shoulder has not decayed at the edge of either window ($e^{-37/\tau_2} = 0.69\text{--}0.93$ for $\tau_2 = 100\text{--}500$ ns), which is why a window-edge normalization would bias $g^{(2)}(0)$ low.

S5.3 Monte-Carlo Bayes reference

For each evaluation acquisition the reference computes exact Poisson likelihoods of the histogram and of the singles count for M sites drawn from the prior and classifies by posterior mass (Sec. S3).

S5.4 Optimization of the measurement setting

The histogram simulator is differentiable with respect to the measurement setting $\theta = (\log s, \log \tau_{\max})$, where s is the excitation saturation parameter and $\tau_{\max}$ the half-width of the correlation window (121 bins). The saturation model scales the detected signal rate and the bunching amplitude as $2s/(1+s)$ (normalized to their values at $s = 1$), the antibunching time as $2/(1+s)$, and the background rate as s . The Poisson counts are reparameterized as $\mu + \sqrt{\mu} \epsilon$ so that pathwise gradients of the task loss reach θ , and the setting and the CNN are optimized jointly; the comparison trains the same CNN at the default setting ($s=1, \tau_{\max}=60.5$ ns). In this experiment the singles-rate input is evaluated at $s = 1$. The SNN and the LM fit were not trained or evaluated at the optimized setting.

S5.5 Experimental data

The experimental data are the sps-quality dataset of Kedziora *et al.*: eight pulsed-excitation (80 MHz) HBT series of an InGaAs/GaAs quantum dot at $1.3 \mu\text{m}$ (FI-SEQR demonstrator), each a delay histogram resolved into 10-s snapshots. One series ($2.5 \mu\text{W}$) was recorded in two sessions, which we keep as separate records, giving nine records and 2016 windows of 30 s. The estimator for these data is a regression network of the same convolutional architecture on a 91-bin window of ± 45.5 ns, trained only on histograms from a pulsed-excitation version of the histogram simulator (a comb of exponential peaks at multiples of the 12.5-ns period, with the zero-delay peak scaled by $g^{(2)}(0)$, side-peak bunching, uncorrelated background, and timing jitter). Its parameter prior spans count rates of 2–18 kcps, which bracket the count rates stated for the dataset, $g^{(2)}(0)$ of 0.02–0.95, and lifetimes of 0.7–4 ns; no experimental histogram is used for training. The auxiliary inputs are $\log T$, $\log$ total counts, and $\log$ counts within 4 ns of zero delay. Reference values are the peak-area estimates of each record’s full accumulation, and the conventional comparison is peak-area analysis of the same 30-s windows (peak areas integrated over ± 3 ns with the inter-peak floor subtracted; the pulse comb is located by folding each record’s full accumulation onto the repetition period). Three records spanning the excitation-power range are held out. Two variants add generic domain randomization (linear tilts and broad spectral bumps) or a residual refiner trained as a WGAN-GP on the six other records to make simulated histograms indistinguishable from experimental ones. The in-simulation error shown in main-text Figure 4(d) is that of a network of the same type, trained in the same way and evaluated on simulated 30-s windows with $g^{(2)}(0)$ between 0.30 and 0.56, the range of the held-out records.

S5.6 Closed-loop triage

Each simulated field presents 48 sites in random order. The per-site probability is supplied by a CNN trained like the variable-exposure CNN but with an additional output for the *good* label (pure, bright, and non-blinking), which is re-evaluated on the accumulated histogram after every exposure step; the SNN was not used in the closed-loop study. The agent observes an eight-dimensional summary (probability, its margin, dwell time, counts, count rate, and the fractions of the queue and time budget used) and chooses among five actions: expose the current site for 0.25, 1, or 4 s; reject it and move on; or certify it and move on. The reward is +1 for a correct certification, −3 for a false one, −1 for rejecting a good emitter, and −0.02 per second of measurement; moving to the next site costs 0.5 s. The agent is a discrete-action soft actor–critic with proportional prioritized experience replay (hyperparameters in Sec. S9); the comparison agent uses uniform replay. The baselines are fixed-dwell rasters with 0.5 to 8 s per site and a threshold controller that exposes a site in 0.25-s steps until the margin $|2p - 1|$ reaches a threshold (or the dwell reaches 5 s). All policies are evaluated on the same 30 held-out fields with the same noise seeds.

S5.7 Transfer between emitter classes

Each emitter class is described by a template graph whose nodes are the electronic states and whose edges carry the ranges of the transition rates, the brightness, and the blinking probability. Two rounds of edge-conditioned message passing are pooled into an eight-dimensional embedding that is added to the estimator inputs. During training every

batch draws new procedurally generated synthetic classes (random but physically consistent templates), so the encoder must learn the relation between the level-structure graph and the photon statistics rather than memorize a few embeddings. The four platform priors (NV, hBN, GaN, SiV) are not used in this training and serve only for evaluation, on simulated data. The comparison includes the same estimator without graph input, trained on the same synthetic classes, a reference model trained directly on the four platform priors, and, for comparison, graph conditioning trained on two platform priors only.

S5.8 Energy estimate

The number of synaptic operations of the SNN is counted on the evaluation set (each input coincidence contributes one operation per first-layer neuron, and each hidden spike one per neuron of the next layer) and multiplied by 23.6 pJ, the minimum energy per synaptic spike operation reported for the 14-nm Loihi chip. The CNN performs a fixed number of multiply-accumulates per decision, priced at 4.6 pJ (32-bit floating point: 3.7 pJ multiply and 0.9 pJ add) or 0.23 pJ (8-bit integer: 0.2 pJ multiply and 0.03 pJ add), the values given by Horowitz for 45-nm CMOS (cited in the main text). The estimate omits memory access and combines values for different process nodes, so it is an order-of-magnitude comparison only.

S6 Estimator training and evaluation

The CNN has convolutional layers with 16, 32, and 32 channels (kernel sizes 7, 5, and 3), average pooling, and fully connected layers of 128 and 64 units, and it receives $\log(1+n)$ histograms together with the five auxiliary inputs listed above. The SNN has hidden layers of 196 and 128 leaky integrate-and-fire neurons with threshold 1, membrane decay factor $\beta = 0.90$, and readout decay factor 0.95; it is trained with a fast-sigmoid surrogate gradient, a loss applied at several slices, and a penalty on the spike rate. Both networks have classification and regression outputs and are trained with Adam (learning rate 10^{-3} with cosine decay, batch size 256, 1500 steps for the CNN) on freshly simulated acquisitions. Tables S9–S12 give the full accuracy and error sweeps, the early decisions of the SNN, and the energy estimate.

Table S9: Balanced classification accuracy (% , mean $\pm$ 1 s.d. over five noise seeds; boundary band excluded) versus acquisition time on the NV prior (long exp.: trained on 30-s acquisitions only; variable exp.: trained on $T \sim \log \mathcal{U}[0.03, 30]$ s). The last row is the Monte-Carlo Bayes reference under the assumed simulation prior with $M = 6000$ reference sites (one reference sample); its dependence on M is given in Table S6.

Estimator	0.03 s	0.1 s	0.3 s	1.0 s	3.0 s	10.0 s	30.0 s
LM fit	59.7 ± 1.3	67.1 ± 0.6	70.4 ± 0.8	80.9 ± 0.7	84.1 ± 0.9	91.1 ± 0.3	93.7 ± 0.6
CNN, long exp.	51.0 ± 0.2	57.3 ± 0.4	65.7 ± 0.5	77.4 ± 0.6	88.6 ± 0.4	95.1 ± 0.5	98.0 ± 0.1
CNN, variable exp.	63.2 ± 1.0	71.6 ± 0.8	81.1 ± 0.6	89.3 ± 0.8	93.9 ± 0.4	97.3 ± 0.4	98.0 ± 0.1
SNN, variable exp.	62.4 ± 0.9	70.2 ± 1.1	79.9 ± 0.7	88.4 ± 0.7	93.2 ± 0.5	97.3 ± 0.4	97.9 ± 0.3
Bayes ref. ($M=6000$)	62.9	71.1	78.0	86.4	91.2	92.7	93.2

Table S10: $g^{(2)}(0)$ regression MAE versus acquisition time (mean $\pm$ 1 s.d. over five seeds; all sites including the boundary band).

Estimator	0.03 s	0.1 s	0.3 s	1.0 s	3.0 s	10.0 s	30.0 s
LM fit	0.475 ± 0.001	0.435 ± 0.024	0.345 ± 0.007	0.247 ± 0.016	0.185 ± 0.001	0.154 ± 0.002	0.141 ± 0.002
CNN, long exp.	0.210 ± 0.000	0.201 ± 0.001	0.183 ± 0.001	0.145 ± 0.001	0.111 ± 0.001	0.077 ± 0.001	0.054 ± 0.000
CNN, variable exp.	0.200 ± 0.001	0.175 ± 0.002	0.147 ± 0.001	0.113 ± 0.001	0.087 ± 0.001	0.066 ± 0.001	0.057 ± 0.000
SNN, variable exp.	0.206 ± 0.002	0.182 ± 0.003	0.156 ± 0.002	0.123 ± 0.002	0.097 ± 0.001	0.078 ± 0.001	0.070 ± 0.001

Table S11: Early decisions of the SNN during a 1-s exposure: median and mean time at which the output first crosses the commitment gate θ (on $|p_1 - p_0|$), balanced accuracy of the committed decisions, and the fraction of sites that never cross the gate within the exposure.

Gate	Median (ms)	Mean (ms)	Bal. acc. (%)	Never-commit
$\theta = 0.5$	312	386	86.2	4.5%
$\theta = 0.6$	375	446	87.1	8.4%
$\theta = 0.8$	500	572	88.4	22.2%
$\theta = 0.9$	625	659	88.2	34.5%
$\theta = 0.95$	812	727	88.2	42.7%

Table S12: Operation-count energy estimate per decision: measured SNN synaptic operations (synops) at 23.6 pJ each (minimum value reported for Loihi) and CNN multiply–accumulates at 4.6 pJ (32-bit floating point) or 0.23 pJ (8-bit integer) each (45-nm CMOS). The last two columns give the SNN energy divided by the CNN energy; values above 1 mean that the SNN estimate is higher.

T (s)	synops (10^3)	SNN (nJ)	CNN FP32 (nJ)	CNN INT8 (nJ)	SNN/CNN FP32	SNN/CNN INT8
0.1	32.6	768	1385	69.3	0.55	11
1.0	177.8	4195	1385	69.3	3.03	61
10.0	1639.4	38689	1385	69.3	27.93	559

S7 Optimization of the measurement setting

The setting is optimized with Adam at learning rate 3×10^{-2} jointly with the CNN (learning rate 10^{-3}), with soft penalties keeping $s \in [1/8, 8]$ and $\tau_{\max} \in [15, 120]$ ns. Table S13 reports balanced accuracy and MAE for both settings. The profile Fisher information for the dip depth D (driven by ρ^2) treats the antibunching time, bunching amplitude, bunching time, and overall scale as unknown, $I_{\text{eff}} = 1/[I^{-1}]_{DD}$, averaged over 400 sites of the NV prior; within the saturation model above it increases monotonically over $1/8 \leq s \leq 8$ (main-text Figure 4(b)), so it does not by itself select $s \approx 1$.

Table S13: Joint optimization of the measurement setting with the CNN (one training run per setting; 2000 evaluation sites): balanced accuracy (%) and $g^{(2)}(0)$ MAE at the default setting ($s=1$, $\tau_{\max}=60.5$ ns) versus the optimized setting ($s^*=1.03$, $\tau_{\max}^*=36.9$ ns).

T (s)	Acc. default	Acc. optimized	MAE default	MAE optimized
0.03	75.8	78.5	0.166	0.162
0.1	81.0	82.1	0.145	0.143
0.3	86.7	86.6	0.125	0.124
1.0	92.9	92.7	0.102	0.106

S8 Experimental-data analysis

Table S14 lists the nine records with their reference values and the split, and Table S15 the errors on the 30-s windows. The WGAN-GP refiner has a residual generator (two hidden layers of 256 units, 16-dimensional latent input) acting on $\log(1+n)$ histograms and a critic with 256 hidden units, a gradient penalty of 10, two critic steps per generator step, and Adam (learning rate 10^{-4} , $\beta_1 = 0.5$, $\beta_2 = 0.9$) for 1500 generator iterations.

Table S14: The eight experimental quantum-dot HBT measurement series (sps-quality, FI-SEQR demonstrator), analyzed as nine records: the 2.5- μ W series was recorded in two sessions (day 1/day 2), kept separate here. $g^{(2)}(0)$ references from the peak-area analysis of each full accumulation.

Series	$g^{(2)}(0)$ ref.	T (s)	30-s windows	Split
10uW 12000cps	0.544	1210	40	held out
10uW 6000cps	0.572	3010	100	train
1p2uW 3000cps time bin width 128 p	0.198	23950	798	train
20uW 7000cps	0.553	4780	159	train
2p5uW 4000cps day1	0.322	4320	144	held out
2p5uW 4000cps day2	0.263	6190	206	train
30uW 7000cps	0.690	4780	159	train
4uW 4100cps	0.334	9390	313	train
8uW 5100cps	0.487	2920	97	held out

Table S15: Mean absolute error of $g^{(2)}(0)$ estimated from 30-s windows of the experimental records, relative to the peak-area value of each record’s full accumulation, for the three held-out records and the six records used to train the refiner.

Method	Held-out series	Train series
Peak-area analysis	0.290	n/a
Simulation only	0.147	0.202
Simulation + domain rand.	0.164	0.219
Simulation + WGAN-GP refiner	0.160	0.212

S9 Closed-loop triage

The soft actor–critic agent has two 128-unit critics and a 128-unit policy, discount 0.98, target smoothing 5×10^{-3} , a learned temperature with target entropy $0.25 \ln |A|$, batch size 256, and one update per two environment steps after 2000 warm-up steps, for 40 000 environment steps in total. The prioritized replay buffer is the proportional variant (sum tree, $\alpha = 0.6$, β annealed from 0.4 to 1, $\epsilon = 10^{-3}$). Table S16 lists every policy on the shared evaluation fields.

S10 Transfer between emitter classes

The graph encoder embeds each template graph (three state nodes with one-hot identity and a radiative flag; four directed transition edges carrying log-rate midpoints, log-range spans, bunching amplitude, log brightness, and blinking probability) with two rounds of edge-conditioned message passing (32 hidden units, gated-recurrent-unit node updates) and mean pooling to an eight-dimensional vector. Table S17 gives the results per platform prior.

S11 Comparison with reported results of related work

Table S18 summarizes the reported results of related approaches, as stated in the respective publications, next to the corresponding results of this work.

Table S16: Closed-loop triage on 30 held-out simulated 48-site fields: measurement time per field (mean $\pm$ s.d. over fields), precision and recall of the certified good emitters, and certified good emitters per minute. The threshold controller exposes a site in 0.25-s steps until $|2p - 1|$ reaches the stated value or the dwell reaches 5 s; the oracle stops each site as soon as the estimate agrees with the ground truth.

Policy	Time (s)	Precision	Recall	Good/min
Threshold controller, $ 2p - 1 \geq 0.8$	78 ± 11	0.884	0.830	6.58
Threshold controller, $ 2p - 1 \geq 0.9$	107 ± 14	0.902	0.873	4.97
Threshold controller, $ 2p - 1 \geq 0.95$	127 ± 14	0.895	0.882	4.25
Fixed-dwell raster, 0.5 s	48 ± 0	0.839	0.724	9.21
Fixed-dwell raster, 1.0 s	72 ± 0	0.843	0.795	6.78
Fixed-dwell raster, 2.0 s	120 ± 0	0.874	0.858	4.37
Fixed-dwell raster, 4.0 s	216 ± 0	0.874	0.869	2.45
Fixed-dwell raster, 8.0 s	408 ± 0	0.898	0.871	1.31
RL agent, prioritized replay, seed 0	244 ± 23	0.886	0.867	2.18
RL agent, prioritized replay, seed 1	259 ± 29	0.875	0.868	2.07
RL agent, uniform replay, seed 0	122 ± 24	0.876	0.833	4.17
RL agent, uniform replay, seed 1	118 ± 21	0.900	0.833	4.35
Oracle stopping (bound)	47 ± 5	0.942	0.904	11.94

Table S17: Cross-platform transfer: balanced accuracy (%) on simulated data from each platform prior at two acquisition times (mean $\pm$ 1 s.d. over three seeds). The four platform priors are not seen in training by the two models trained on synthetic classes.

Model	NV 0.3s	NV 3.0s	hBN 0.3s	hBN 3.0s	GaN 0.3s	GaN 3.0s	SiV 0.3s	SiV 3.0s
Uncond. (synth.)	73.3 $\pm$ 0.4	88.6 $\pm$ 0.6	75.8 $\pm$ 1.0	85.7 $\pm$ 0.8	65.4 $\pm$ 0.3	75.5 $\pm$ 0.7	66.3 $\pm$ 0.9	81.2 $\pm$ 0.9
Graph-cond. (synth.)	75.9 $\pm$ 0.1	90.4 $\pm$ 0.6	74.8 $\pm$ 0.6	84.8 $\pm$ 0.2	70.6 $\pm$ 0.5	84.1 $\pm$ 0.4	70.7 $\pm$ 0.2	89.8 $\pm$ 1.1
Reference (four priors)	72.4 $\pm$ 0.6	86.2 $\pm$ 0.2	74.7 $\pm$ 0.7	84.4 $\pm$ 0.2	73.6 $\pm$ 0.5	89.7 $\pm$ 0.3	72.7 $\pm$ 0.5	89.9 $\pm$ 0.9

Table S18: Reported results of related machine-learning and automated approaches to single-photon-emitter characterization, as stated by the respective authors (references as in the main text), and the corresponding results of this work. Tasks, data, and baselines differ between entries, so the reported numbers are not directly comparable.

Work	Data	Method	Reported result
Kudyshev <i>et al.</i> , 2020	Sparse HBT autocorrelation histograms	Supervised classification (single or not single), including a 1D convolutional network	95% accuracy within an integration time of less than 1 s; roughly 100-fold speedup over Levenberg–Marquardt fitting
Kudyshev <i>et al.</i> , 2023	Photon-correlation histograms in antibunching (quantum super-resolution) microscopy	Convolutional-network regression	12-fold speed-up compared with fitting-based autocorrelation measurements
Narun <i>et al.</i> , 2022	Photoluminescence images	Image analysis and regression fitting	Automated identification and classification of emitters in nanodiamond arrays and hBN flakes
Xu <i>et al.</i> , 2024	Higher-order photon-correlation data	Two-dimensional convolutional-network classification of photon states up to $ 3\rangle$	94% overall accuracy; 90% with 800 co-detection events
Kedziora <i>et al.</i> , 2023	Eight experimental HBT datasets of one InGaAs/GaAs quantum dot	Bootstrap data augmentation to quantify the uncertainty of early estimates	Significant uncertainty from Poisson variability; least-squares fitting comparable to Poisson likelihood
Kedziora <i>et al.</i> , 2025	Same quantum dot, eight excitation contexts	Transfer learning with linear and ensemble regressors (trained on seven contexts, tested on the eighth)	Regressors can outperform least-squares fitting within trained contexts; success of transfer less assured
This work	Simulated CW HBT histograms and time-sliced coincidence records; experimental data of Kedziora <i>et al.</i> for testing	Spiking and convolutional networks; RL exposure control; graph conditioning on the emitter class	In simulation: $6\times$ less acquisition time than a multi-start LM fit at 90% balanced accuracy, and $1.7\times$ faster field screening than fixed dwell. On experimental 30-s windows: $2.0\times$ lower $g^{(2)}(0)$ error than peak-area analysis